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ENGG1120F Quiz 2

Due: 2-Feb-2026, 23:59 HKT.

INSTRUCTIONS

1. Answer all **3 Questions**. Total 30 marks.
2. Please **answer on the hard-copy**, then scan and submit online through Gradescope.
3. Due time: 2-Feb-2026, 23:59 HKT. No late submissions will be accepted.
4. By submitting this assignment, you are assumed to have read and understood the course assessment policy, which you can find on Blackboard. You are also assumed to have read and understood CUHK's guideline on academic honesty and on the use of AI tools.

Question 1. (10 marks)

Find the solution set for the following system of linear equations

$$\begin{cases} x_1 + 3x_2 + 5x_3 = 7 \\ x_1 + 4x_2 + 7x_3 = 10 \\ 8x_2 + 16x_3 = 34 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & 7 \\ 1 & 4 & 7 & 10 \\ 0 & 8 & 16 & 34 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 8 & 16 & 34 \end{array}\right] \xrightarrow{R_3 \to R_3 - 8R_2} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 10 \end{array}\right]$$

solution set: $\emptyset$

Question 2. (10 marks)

Determine the solution set of the following homogeneous system

$$\begin{cases} x_1 + 3x_2 + 5x_3 + 7x_4 = 0 \\ x_2 + 2x_3 + 3x_4 = 0 \\ 8x_2 + 16x_3 + 34x_4 = 0 \end{cases}$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 8 & 16 & 34 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 8R_2} \begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

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let $x_3 = 1$. $\begin{cases} x_1 + 3x_2 + 5x_3 + 7x_4 = 0 \\ x_2 + 2x_3 + 3x_4 = 0 \\ 10x_4 = 0 \end{cases}$

solution set: $k\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}$, $k \in \mathbb{R}$

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$-2-(k-12) = -k+10$

Question 3. (10 marks)

Let $k \in R$ be a real parameter and consider the matrix $A = \begin{bmatrix} -1 & 1 & -4 & 0 \\ 3 & -4 & k & 4 \\ 2 & -3 & 6 & 4 \end{bmatrix}$. Determine all value(s) of k for which rank$(A) = 3$.

$$\begin{bmatrix} -1 & 1 & -4 & 0 \\ 3 & -4 & k & 4 \\ 2 & -3 & 6 & 4 \end{bmatrix} \xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 + 3R_1} \begin{bmatrix} -1 & 1 & -4 & 0 \\ 0 & -1 & k-12 & 4 \\ 0 & -1 & -2 & 4 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} -1 & 1 & -4 & 0 \\ 0 & -1 & k-12 & 4 \\ 0 & 0 & -k+10 & 0 \end{bmatrix}$$

For rank to be 3, there should be 3 pivots.

Namely $-k+10 \neq 0 \implies k \neq 10$